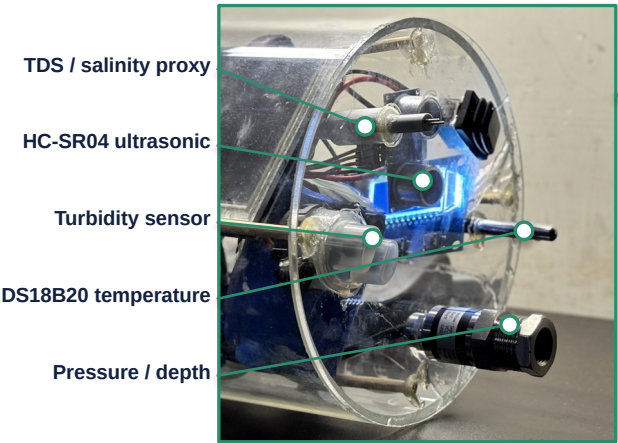


Your
Team
Name

TECHNICAL APPROACH



SENSE · REAL ENVIRONMENT



ADC · 1-Wire · I²C

also on board: IMU + magnetometer · BMP390 cabin pressure · INA219 ×2 (mounted, not yet commissioned)

ONE ADAPTATION POLICY · TWO INPUTS

A0 sensor
TDS_PROXY · ADC

Web command
console → UART

TECHNOLOGIES TO BE USED
STM32F446RE (Cortex-M4F) · embedded C / C++
TIM6 + DMA + 12-bit DAC · ADC / 1-Wire / I²C
Sallen-Key LPF + TLV9062 op-amp (analog)
Node.js + React console · Python / NumPy · OpenSCAD

STM32F446RE · EMBEDDED CONTROL *methodology & process*

THE NEXT PING IS COMPUTED ON THE MCU

01

ACQUIRE
ADC · 1-Wire · I²C · stale / out-of-range readings rejected

02

DECIDE
smoothing + hysteresis → CLEAR / TRANSITION / MURKY

03

SYNTHESISE
LFM chirp · geometric sweep · Barker-13 × Hann / Hamming / Blackman

04

STREAM
TIM6 → DMA → 12-bit DAC1 (PA4) · 1 MS/s · CPU sleeps between pings

06

SELF-VALIDATE
ADC3 + DMA capture of the DAC pin (PA4 → A5) → FFT / spectral analysis

MCU ADC CAPTURE 42.00 kHz centre · 4 kHz sweep · 2 ms pulse · HANN
LINK + STREAM 330 / 330 commands acknowledged · 0 DMA underruns

ADAPTED NEXT PING

	CLEAR	TRANSITION	MURKY
Centre	42 kHz	40 kHz	38 kHz
Bandwidth	4 kHz	2 kHz	1 kHz
Pulse	2 ms	4 ms	8 ms
Amplitude	35 %	50 %	62 %

CLEAR → shorter, sharper ping MURKY → lower band · longer · more output

05

CONDITION · analog front end

2nd-order LPF
70.7 kHz cut-off
TLV9062 ×1.5

DESIGNED · NEXT BUILD

AQUASDR PAYLOAD physical prototype · acrylic payload hull
External scope validation → PENDING · AUV hull-slot pod → NEXT BUILD

built / running

designed / next build

SENSE THE WATER → COMPUTE THE NEXT PING → STREAM THE WAVEFORM → VERIFY THE OUTPUT